

Hack The World(s)

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Agenda

- 1 What You Have & What We Expect
- 2 Presentation Requirements
- 3 The Maze Example
- 4 Timeline & Judging

The Data You Will Work With

Two main types of data are available for the challenge.

1 — Datasets we provide

Ready-to-use datasets from different domains:

- **Neuroscience** — neural recordings
- **Audio** — audio/speech files
- **Physion** — physical reasoning / object dynamics

Just load them and start training.

2 — Data you generate

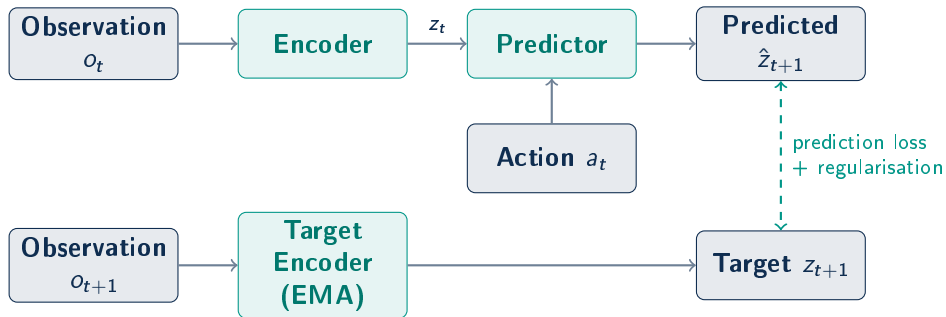
Datasets you build yourselves, following our example:

- **Two rooms**
- **Maze** — our reference example (see Section 2)
- Serves as a **pipeline** you can reuse and adapt

Full generation code is provided as a starting point.

You may use one type, the other, or both.

The Model — A JEPA Baseline

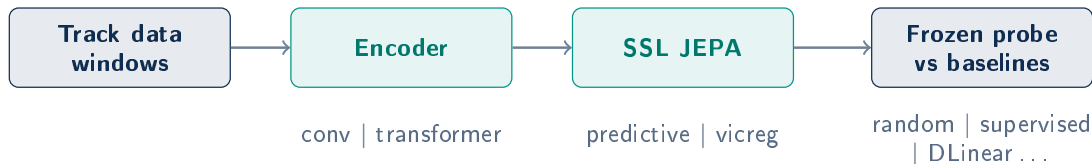


What we provide

A working **JEPA** that predicts *in representation space*, with regularisation losses to prevent collapse. Three tracks: **image**, **video**, **action-conditioned video**.

Pipeline — a modular, config-driven JEPA scaffold

One generic engine; each track only fills a small #TODO surface.



Provided — don't touch

- Encoders: Conv1d / Transformer
- Objectives: predictive JEPA / VICReg
- Predictor: RNN (GRU) / Transformer
- AMP trainer + probe harness + metrics

You fill — the #TODO surface

- `build_pretrain_dataset` — SSL windows
- `build_eval_dataset` — (x, y) pairs
- `task_type`, `in_channels`
- *optional: encoder / metric / augment*

One Pipeline, Many Tracks

Same engine for every modality — a track = data loaders + a task declaration.

Track	SSL objective	Downstream	You write
Neuroscience (EEG)	vicreg (two-view)	classification probe	loader + augment
Audio / speech	vicreg (two-view)	classification probe	loader + augment
Physion (video)	predictive (action-cond.)	forecasting / probe	loader
Maze / Two-rooms (gen.)	predictive (action-cond.)	planning / probe	loader
Time series (fin. / LTSF)	predictive	forecasting / regr.	loader + baseline

Pick *predictive* when...

- there is temporal / sequence structure to predict
- an action or control channel is available
- $\Rightarrow$ video, planning, time series

Pick *vicreg* when...

- you have strong, meaning-preserving augmentations
- no natural “next step” to predict
- $\Rightarrow$ EEG, audio, point clouds

What We Expect From You

Start from a baseline example, then make it yours.

01 A working model

Train a JEPA world model on the data of your choice.

- Keep it reproducible (`config.yaml`)
- Save at least one stable checkpoint

02 10-minute presentation

Walk us through your approach, experiments, and findings.

- Explain your architectural choices
- Show what the model learned

The Gold Standard

What makes a great submission?

- **Clear problem framing**
- **Quantitative results** vs. baseline
- **A genuine insight** or finding
- **Honest discussion** of limits
- **At least one ablation** study



What Your Presentation Should Cover

Your 10-minute presentation should address the points below.

Data → **Architecture** → **Training** → **Inference** → **Evaluation**

(plus optional **Bonuses** to go further)

Explain the data you trained on.

1 Where is it from?

- **Generated** by you
- **Chosen** from the provided datasets
- **Found elsewhere** (cite the source)

Modality, size, difficulty?

2 What does it look like?

Show it — stats, samples, a projection (PCA):



3 How did you prepare it?

- Normalisation
- Encoding / decoding target
- Data augmentation?

Any preprocessing worth noting?

Describe your model and how it was trained.

The model

- **Unchanged?** A quick JEPA recap
- **Modified?** Why, what, intuition, papers
- Number of parameters & modules

Loss & training dynamics

- Which (sub)losses? how balanced?
- Representation collapse — observed?
- Regularisation & stability



How did you train, and how did you iterate efficiently?

Setup

- Batch size, optimiser
- Learning-rate scheduler
- Number of epochs

Comparing without full training

- A **proxy metric** to rank runs early
- A strategy to compare *without* a full run
e.g. scaling laws, ablations, ...

How do you use the trained model?

Inference strategy

- Method → Mode 1 / Mode 2, MCTS, actor, ...?
- Performance vs. inference time
- Any inference tricks?

Performance vs. time



more compute, better rollout?

How good is it, and how do you know?

Robustness

- Ablations
- Scaling laws
- Seed & training stability

Performance

- Dataset performance
- Compare to a baseline

Reach

- Application to real-life situations
- A step towards AGI?

Optional — ways to stand out.

Rewarded

- High performance
- Hyperparameter tuning
- Extensive training-stability analysis — esp. visualising representation collapse

Going further

- Handmade (non-synthetic) dataset
- Method that scales to other datasets of the field (e.g. time series)
- Hierarchical world models

The Maze Example

Schedule

17:30 Code submission

18:00 Slides submission & pre-jury

19:00 Final jury (*qualified teams*)



Judging Criteria

- **Technical quality**

Robustness, code cleanliness & architecture

- **Results**

Performance, metrics & meaningful comparisons

- **Presentation**

Clarity, storytelling & visual aids

- **Originality**

Ablations, novel insights & going the extra mile

Now go build your world.

Good luck, and have fun!